

GATESVILLE, TX, USA

WMO: 722323

Lat: 31.421N

Lon: 97.797W

Elev: 906

StdP: 14.22

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 00362

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.1	28.2	9.5	9.2	35.5	14.1	11.6	38.3	23.5	56.3	20.5	58.3	3.0	10	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.9	100.3	73.3	99.3	73.6	97.2	73.6	78.0	89.1	77.2	89.0	76.4	87.9	7.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	136.9	79.7	74.6	134.0	79.6	73.2	127.8	79.3	41.8	90.0	41.2	88.1	40.4	86.7	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.1	17.1	15.5	DB	18.8	104.8	4.8	3.0	15.3	106.9	12.5	108.7	9.8	110.4	6.3	112.6	(4)
			WB	17.0	79.9	4.3	3.2	13.9	82.2	11.4	84.0	9.0	85.8	6.0	88.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	67.5	48.2	52.5	60.0	67.0	74.1	82.3	84.7	85.3	78.7	68.0	58.0	49.7	(6)
		DBStd	15.24	9.77	10.74	9.24	7.30	6.69	3.89	3.53	3.94	5.54	8.52	9.45	9.53	(7)
		HDD50	426	151	90	23	1	0	0	0	0	3	32	126		(8)
		HDD65	1976	528	366	207	63	14	0	0	1	69	246	482		(9)
		CDD50	6797	94	159	333	511	747	970	1075	1096	861	562	272	117	(10)
		CDD65	2871	6	15	51	123	296	520	610	631	412	163	36	8	(11)
		CDH74	33346	54	136	403	1075	2786	6362	7838	8298	4495	1576	264	59	(12)
		CDH80	16820	5	26	83	303	1107	3339	4367	4765	2206	560	56	3	(13)
(14)	Wind	WSAvg	5.8	5.9	6.6	7.2	7.2	6.7	6.2	5.4	4.5	4.1	5.0	5.5	5.4	(14)
(15)	Precipitation	PrecAvg	33.3	2.0	2.7	3.1	2.6	4.1	4.1	2.1	2.4	2.4	3.3	3.0	2.2	(15)
(16)		PrecMax	50.3	4.5	7.1	10.6	5.9	9.8	9.4	7.9	8.9	4.8	6.1	10.9	6.6	(16)
(17)		PrecMin	22.6	0.4	0.0	0.9	0.7	1.2	1.1	0.0	0.1	0.6	0.4	0.4	0.2	(17)
(18)		PrecStd	7.1	1.2	1.9	2.2	1.7	2.4	2.2	1.9	2.4	1.3	1.9	2.6	1.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	80.7	82.4	86.1	91.1	96.7	100.5	102.4	104.2	100.2	92.6	85.8	80.0		(19)
(20)		0.4% MCWB	58.8	61.1	64.0	69.1	71.9	73.4	72.4	72.6	71.3	70.0	69.6	64.2		(20)
(21)		2% DB	73.4	78.8	81.9	86.4	92.5	98.9	100.0	100.5	97.0	89.9	81.1	74.8		(21)
(22)		2% MCWB	58.7	60.7	64.2	67.4	72.9	73.3	73.4	73.6	72.5	70.0	65.6	62.0		(22)
(23)		5% DB	70.2	73.2	79.1	82.5	90.0	96.7	98.9	99.4	93.2	86.1	76.7	70.5		(23)
(24)		5% MCWB	59.1	60.1	63.8	67.0	72.5	73.6	73.7	74.0	72.3	69.4	64.1	60.4		(24)
(25)		10% DB	65.6	70.2	74.8	80.9	86.2	93.1	96.8	97.3	90.8	82.2	72.9	66.1		(25)
(26)		10% MCWB	56.2	60.3	61.9	66.2	71.9	73.3	74.0	74.0	72.1	66.9	63.4	58.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	68.0	68.4	71.0	74.2	79.4	78.3	78.4	78.4	78.0	76.0	72.3	69.0		(27)
(28)		0.4% MCDB	71.5	74.9	78.5	83.0	83.0	89.9	89.3	89.7	88.7	83.6	78.4	73.4		(28)
(29)		2% WB	64.3	66.5	68.5	71.9	76.1	77.2	77.5	77.8	76.7	74.2	70.4	66.5		(29)
(30)		2% MCDB	69.1	71.8	75.2	80.6	85.8	89.2	89.7	89.8	87.0	81.5	75.9	71.7		(30)
(31)		5% WB	61.2	64.4	67.2	70.6	74.9	76.2	76.8	77.1	75.7	72.9	68.1	63.7		(31)
(32)		5% MCDB	66.3	69.2	74.4	79.0	84.1	87.8	88.7	89.3	85.2	80.1	73.6	67.2		(32)
(33)		10% WB	56.9	61.9	65.8	69.0	73.7	75.3	76.0	76.2	74.8	71.5	66.1	60.2		(33)
(34)		10% MCDB	64.5	67.6	71.8	76.9	82.6	86.7	87.5	88.5	84.0	78.7	71.3	65.4		(34)
(35)	Mean Daily Temperature Range	MDBR	22.6	21.9	22.5	22.2	20.2	21.2	21.3	21.9	21.6	23.1	22.7	21.8		(35)
(36)		5% DB MCDBR	29.0	27.1	25.2	24.5	23.6	24.4	24.4	24.4	24.4	24.6	24.8	27.5		(36)
(37)		5% DB MCWBR	16.9	15.2	12.5	11.5	8.4	5.8	5.3	5.7	7.1	10.0	13.0	16.9		(37)
(38)		5% WB MCDBR	22.8	21.7	19.4	19.7	18.9	19.4	19.6	20.6	19.4	18.8	19.1	21.6		(38)
(39)		5% WB MCWBR	16.7	15.0	11.5	11.2	8.9	6.1	5.6	5.6	6.2	9.9	13.1	16.9		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.312	0.325	0.352	0.392	0.423	0.424	0.436	0.427	0.396	0.367	0.339	0.317		(40)
(41)		taud	2.515	2.482	2.421	2.334	2.304	2.341	2.345	2.353	2.386	2.453	2.473	2.505		(41)
(42)		Ebn at Noon	292	297	295	285	276	274	270	272	277	279	278	283		(42)
(43)		Edh at Noon	27	31	36	40	42	40	40	39	36	32	29	26		(43)
(44)	All-Sky Solar Radiation	RadAvg	958	1125	1456	1753	1856	2114	2149	1985	1660	1327	1007	846		(44)
(45)		RadStd	102	182	126	162	174	168	161	121	141	198	124	116		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+131

Nomenclature: See separate page